Information Engineering and Technology Faculty
German University in Cairo

AI Driven Predictive Maintenance

Bachelor Thesis

Author: Abdelrahman Ibrahim Fouad
Supervisor: Amr Talaat Abdel-Hamid
Reviewer: Hossam Eldin Hassan
Submission Date: 19 May, 2024

This is to certify that:

(i) the thesis comprises only my original work towards the Bachelor Degree

(ii) due acknowledgement has been made in the text to all other material used

Abdelrahman Ibrahim Fouad

19 May, 2024

Abstract

This thesis involves a detailed analysis of the Remaining Useful Life (RUL) of turbofan engines and NASA Li-ion battery datasets. The research aims to using the traditional machine learning algorithms and deep learning algorithms to forecast the risk of failure.

The dataset is preprocessed using correlation analysis, feature engineering, and scaling among other steps, in order to ensure that high quality input data is fed into the models.

The study consists of three rounds of experiments for the Nasa turbofan dataset which each involve different features selection to evaluate how this impacts model performance. Additionally, time series analysis is done on the dataset for the purpose of capturing temporal patterns that exist in it thus ensuring more accurate RUL estimation. Therefore, main aim is to compare performance between different modeling approaches and identify the best algorithm for risk prediction.

Furthermore, experimental results are thoroughly analyzed and compared across various modeling techniques employed. These findings provide insights on strengths as well as limitations of traditional machine learning algorithms and deep learning algorithms for RUL prognostics. Further still, importance of feature engineering and temporal analysis in improving accuracy with respect to RUL predictions has been brought out by this study.

The findings of this research enrich the knowledge on AI-guided prognostics for lithium-ion batteries and turbofan engines in the domain of predictive maintenance

and the use of the IoT devices in the industrial development. They will help decision making regarding maintenance planning and resource allocation which would lead to higher reliability, less down time and cost savings. Altogether, this thesis presents an all-round examination of the actual research carried out, its methodology as well as insights obtained from the comparative study of traditional machine learning algorithms and deep learning algorithms for RUL prognostics.

This information is key in setting a basis for better strategic planning by future scholars towards better predictive maintenance techniques in critical industrial applications.

Contents

Chapter 1 Introduction …… 1

1.1 Motivation…….. 1

1.2 Aim of the project…….. 3

Chapter 2 Background Material ……. 6

Chapter 3 Methodology ……10

3.1 Implementation Details ……... 10

3.1.1 Datasets description …… 10

3.1.2 Data preprocessing and Correlation analysis…… 11

3.1.3 Traditional machine learning techniques…… 15

3.1.4 Deep machine learning techniques…… 18

3.2 Experimental Results …….…. 21

3.3 Module implementation……... 25

3.4 Android application……. 28

Chapter 4 Conclusion ……...…... 31

References ……. 36

List of Figures

Figure 1: Nasa turbo fan photo……5

Figure 2: Ai use in industrial.….……..……5

Figure 3: Nasa LI-ion battery……..….5

Figure 4: Ai driven predictive Machine learning approaches…….9

Figure 5: Nasa turbofan correlation analysis dataset…… 13

Figure 6: LI-ion battery correlation analysis dataset…… 13

Figure 7: Min-Max scaling for Nasa turbofan dataset……...14

Figure 8: LSTM model prediction vs actual graph for Nasa turbofan dataset…….…….24

Figure 9: LSTM model reduced prediction vs actual for Nasa turbofan dataset ….……24

Figure 10: Random Forest model prediction vs actual graph for LI-ion battery dataset……...24

Figure 11: Esp8266 Module…….….26

Figure 12-14: Android Ide screenshots for the module prediction……27

Figure 15-26: Android application screenshots for prediction application …….....29

List of Tables

Table 1 Authors results before for the C-MAPPS dataset ……... 8

Table 2 Nasa turbofan dataset description……..... 10

Table 3 Nasa Li-ion battery dataset……..... 11

Table 4 Machine learning models results……..……... 21

List of Terms

AI	Artificial Intelligence
C-Mapps	Nasa turbo fan dataset
e.g.	Example
FD001	Subset of C-Mapps Dataset
LI-ion	Lithium-ion battery
LSTM	Long short-term memory
NN	Neural Network
RELU	Rectified Linear unit
RUL	Remaining useful life
RNN	Recurrent neural network
R2	Pearson correlation Coefficient
SVR	Support Vector regression
SVM	Support Vector Machine
XGBOOST	Extreme Gradient Boosting

List of Symbols

Scaled	Scaled value
X	Value we will scale by min-max scaling
X MIN	Minimum row value
X MAX	Maximum row value

ACKNOWLEDGMENTS

It is with immense gratitude that I acknowledge the support and help of my supervisor Professor Amr Talaat who believed in me and was continuously supporting me in every problem that occurred during this research and project. Our cooperation was truly an inspiring experience, his comments were of critical importance, patience and hard work that made this thesis possible. Very special thanks to doctor Samar Shoukry for her eagerness and great assistance. I would like to dedicate this work to my grandmother Nawal Elbadry, beloved parents, Ibrahim Fouad, Iman Khattab and my sister, Habiba ibrahim. Without your love and support I could not have made it until the end.

Chapter 1

Introduction

1.1 Motivation

Predictive maintenance, empowered by Artificial Intelligence (AI) techniques, has gained significant importance in various industries [Figure 2]. The application of AI in predictive maintenance can lead to substantial improvements in operational efficiency, cost savings, and increased equipment reliability.

Traditional maintenance practices often follow fixed schedules or reactive approaches, leading to unnecessary maintenance activities or unexpected failures. AI-driven predictive maintenance enables condition-based maintenance, where maintenance actions are scheduled based on the actual condition and health of the equipment. This approach can lead to significant cost savings by avoiding unnecessary maintenance and reducing downtime. for example, Airbus, a leading aircraft manufacturer, implemented an AI-driven predictive maintenance system for their A380 aircraft fleet. The system analyzed sensor data, flight data, and maintenance records to predict component failures and recommend optimized maintenance schedules. This implementation resulted in a 30% reduction in maintenance costs and a 25% reduction in unscheduled maintenance events. Unplanned equipment downtime can have severe

financial consequences for industries, causing production losses and delays. AI-driven predictive maintenance helps identify early signs of equipment degradation or failure, allowing proactive maintenance interventions to be performed before a breakdown occurs.

For example, General Electric implemented an AI-based predictive maintenance solution for their gas turbines. By analyzing real-time sensor data, the system detected anomalies and predicted impending failures. As a result, General Electric achieved a 25% reduction in unplanned downtime, saving millions of dollars per year.

AI algorithms can analyze complex datasets, including sensor data, historical maintenance records, and environmental conditions, to identify patterns and correlations that might indicate potential equipment failures. This enables proactive maintenance actions, such as component replacement or repairs, to prevent major breakdowns and extend equipment lifespan.

For example, In the oil and gas industry, AI-driven predictive maintenance has been employed to monitor the health of drilling equipment. By analyzing sensor data, the system detects signs of wear, corrosion, or abnormal operating conditions, allowing operators to schedule maintenance activities before critical failures occur. This approach has led to a significant reduction in equipment downtime and increased overall reliability.

Predictive maintenance can contribute to improved safety and product quality. By identifying and resolving potential equipment issues before they lead to failures, companies can ensure the reliability and consistency of their products, reducing the risk of accidents and defects.

For example, In the automotive industry, AI-powered predictive maintenance systems are used to monitor the health of production equipment, such as robotic assembly lines. By detecting anomalies or deviations in equipment performance, maintenance teams can intervene promptly, preventing quality issues and ensuring the safety of workers.

This study focuses on the application of AI-driven predictive maintenance techniques using the Turbofan NASA dataset [Figure 1] and the Li-ion battery [Figure 3] dataset for Remaining Useful Life (RUL) prediction. the Turbofan NASA dataset provides a valuable resource for studying the degradation patterns of turbofan engines. Understanding the RUL of these engines is of utmost importance to ensure optimal performance, prevent unexpected failures, and reduce maintenance costs. Similarly, the Li-ion battery dataset offers insights into the degradation behavior of batteries, enabling accurate RUL estimation for critical energy storage applications.

1.2Aim of the project

The general aim of this study is to advance the comprehension and accuracy of AI-driven predictive maintenance, particularly focusing on Remaining Useful Life (RUL) estimation. This is pursued through specific objectives, which encompass the exploration of the Turbofan NASA dataset, involving sensor readings analysis, feature identification, and feature engineering to discern underlying patterns. Subsequently, RUL prediction models are developed and evaluated for turbofan engines, including Long Short-Term Memory (LSTM) networks, with performance

comparison. Additionally, the Li-ion battery dataset is scrutinized to comprehend degradation patterns and performance metrics, employing feature engineering techniques. Tailored machine learning models, integrating time series analysis, are then employed for RUL prediction of Li-ion batteries, with subsequent comparison of model performance. Finally, the study extends to deploying the developed machine learning models into IoT devices, with a focus on enhancing accuracy to enable effective predictive maintenance strategies.

This study contributes to the broader field of AI-driven predictive maintenance by exploring advanced machine learning algorithms (e.g. NN, LSTM, XGboost, SVR, and Linear Regression) and highlighting their effectiveness in specific application domains. Motivated by the growing demand for reliable and efficient maintenance strategies, this thesis aims to provide valuable insights into AI-driven predictive maintenance for RUL prediction. The subsequent chapters will focus on methodology, data analysis, modeling techniques, and experimental results, with the ultimate goal of advancing predictive maintenance practices in various industrial domains.

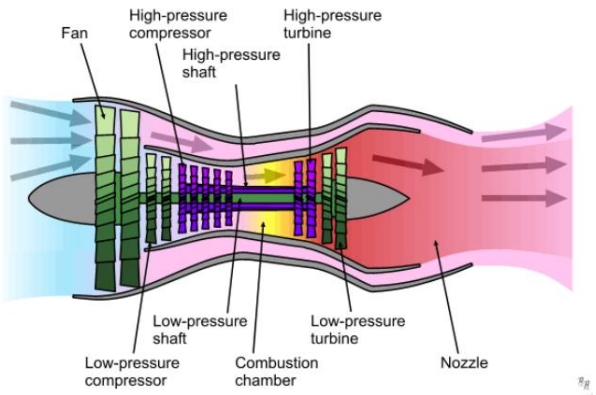

Figure 1

Figure 2

Figure 3

Chapter 2

Background Material

Previous studies have demonstrated the effectiveness of AI techniques in predictive maintenance. Machine learning algorithms, such as support vector machines (SVMs), random forests, and artificial neural networks, have been widely employed for fault detection, diagnosis, and RUL prediction. These approaches leverage historical data to learn patterns and make accurate predictions. However, the majority of research has focused on general predictive maintenance applications, with limited studies specifically targeting turbofan engines and Li-ion batteries. Several investigations have explored RUL prediction for turbofan engines. where researchers from various disciplines like artificial intelligence and statistics apply and test their methodical approaches. The majority of studies, however, only evaluate the overall solution against a final prediction score [Table 1], where we argue that a more fine-grained consideration is required distinguishing between detailed method components to measure their particular impact along the prognostic development process. Feature engineering techniques, such as statistical analysis and time domain features, have been used to extract relevant information from sensor readings and operational parameters. Traditional machine learning methods, including regression models and decision trees, have been applied to estimate RUL. However, these approaches [Figure 4] often struggle to capture the complex degradation patterns and temporal dependencies present in turbofan engine data. Despite the existing research efforts, several

knowledge gaps can be identified: Limited focus on turbofan engines and Li-ion batteries: While AI-driven predictive maintenance has been extensively studied, there is a lack of specific research targeting RUL prediction in turbofan engines and Li-ion batteries. These domains require tailored approaches to capture the unique degradation patterns and performance characteristics. Insufficient consideration of temporal dependencies: The majority of existing studies rely on static feature analysis and overlook the temporal nature of degradation processes. Turbofan engines and Li-ion batteries exhibit dynamic behaviors that evolve over time, necessitating the incorporation of time series analysis and advanced AI algorithms capable of capturing temporal dependencies.

Limited exploration of advanced AI techniques: While traditional machine learning algorithms have been applied in RUL prediction, more advanced techniques, such as LSTM networks and deep learning architectures, have not been extensively explored. These advanced algorithms have the potential to improve RUL estimation accuracy by capturing intricate degradation patterns and temporal relationships. The current research aims to address these knowledge gaps by applying advanced AI techniques to the Turbofan NASA dataset and the Li-ion battery dataset. By leveraging the temporal dependencies and employing state-of-the-art machine learning algorithms. Through a comprehensive evaluation of the existing literature, it becomes evident that there is a need for further research specifically focused on RUL prediction in the context of turbofan engines and Li-ion batteries. The subsequent sections will detail the methodology, data analysis, and modeling techniques employed to fill the identified gaps and advance the field of AI-driven predictive maintenance.

Authors	Year	Technique used	Dataset	Results Achieved
Babu et al. [1]	2016	Deep CNN	C-MAPSS	RMSE for FD002(30.29) RMSE for FD003(19.81)
Zhang et al. [2]	2016	Multi-objective deep belief networks ensemble	C-MAPSS	RMSE for FD002(25.05) RMSE for FD003(12.51)
Malhotra et al. [3]	2020	LSTM-ED	C-MAPSS	The authors have failed to report the RMSE value for each subset of data
Li, Ding, and Sun,[4]	2018	DCNN	C-MAPSS	RMSE for FD002(22.36) RMSE for FD003(12.64)
Zheng et al. [5]	2017	DEEP LSTM	C-MAPSS	RMSE for FD002(24.49) RMSE for FD003(16.18)
Song et al. [6]	2019	Autoencoder-BLSTM	C-MAPSS	The model was only tested on a simpler subset of data called FD001 where the RMSE achieved is 13.63
Wen et al. [7]	2018	ResCNN	C-MAPSS	The authors have failed to report the RMSE value for each subset of data
Li, Li, and He [8]	2019	CNN combined with LSTM	C-MAPSS	RMSE for FD002(20.34) RMSE for FD003(12.46)

Muneer et al. [9]	2021	Attention-based DCNN	C-MAPSS	RMSE for FD002(18.34) RMSE for FD003(13.08)
Zhang et al. [10]	2019	CNN-XGB	C-MAPSS	RMSE for FD002(18.34) RMSE for FD003(13.08)
Ji et al. [11]	2020	PCA-BLSTM	C-MAPSS	The model was only tested on a simpler subset of data called FD003 where the RMSE achieved is 11.1
Wang et al. [12]	2021	MS-CNN	C-MAPSS	The MSE obtained with average test loss is (35.48) The R2 achieved with average test loss id (0.64)

Table 1

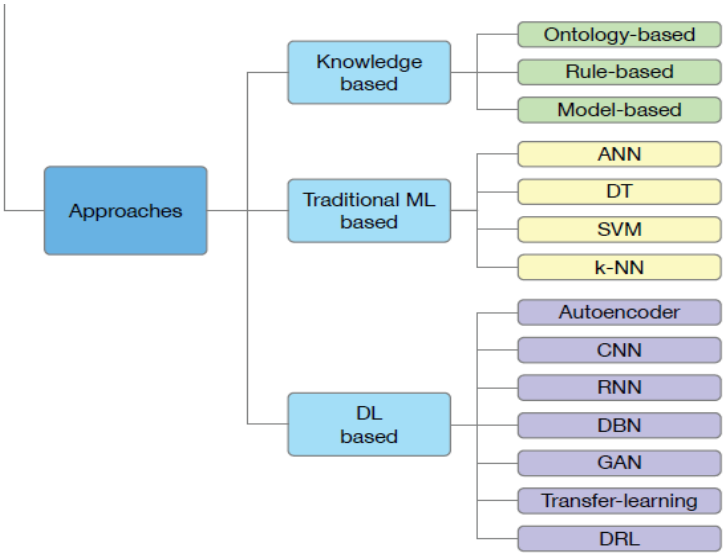

Figure 4

Chapter 3
Methodology
3.1 Implementation Details
3.1.1 Data description

For the First dataset Nasa Turbofan, each column represents a different variable, and each row represents a snapshot of data received during a single cycle of operation. Each engine is equipped with 21 different types of sensors (e.g., pressure, temperature, and shaft speed) that collect data on various engine parameters. C-MAPSS is made up of two datasets: training and testing. Only the first sub-dataset, FD001, is considered in this paper. The first two columns provide the unit number and operating duration in cycles, while the third, fourth, and fifth columns contain operational parameters. As seen in [Table 2], the rest of the columns are sensor measurements.

Index	Data Description
1	Unit number
2	Time cycles
3	Operational setting 1
4	Operational setting 2
5	Operational setting 3
6-26	Sensor measurements 1-21
27	Rul

Table 2

Secondly, The Li-ion battery dataset, each column represents a different variable, and each row represents a snapshot of data measured during a single cycle of operation (e.g., voltage, current, and Temperature) that collect data from various batteries. As seen in [Table 3]

Index	Data Description
1	Voltage measured
2	Current measured
3	Temperature measured
4	Current load
5	Voltage load
6	Time

Table 3

3.1.2 Data preprocessing and Correlation Analysis

A min-max scaling is used for both the training and testing data of the

C-MAPSS and Li-ion battery datasets along each variable to standardize the data and prevent overfitting the model over variables with a high order of magnitude [13,14]. The scaled value X scaled of a vector X is evaluated using the equation. Each column function's lowest and maximum values are represented by Xmin and Xmax, respectively. [Figure 7]

$$\text{Scaled} = (X-X_{min}) / (X_{max}-X_{min})$$

Correlation Analysis, is a method used for feature selection to measure the relationship between variables in a dataset. By calculating the correlation

coefficients. This analysis helped us to identify the dependencies between different features in our datasets.

To assess the correlation between sensor data, a correlation matrix heat map was employed. This visualization tool represents correlation coefficients using both numerical values and color coding. In [Figures 5,6], the correlation matrix heat maps for the C-MAPSS and Li-ion battery datasets are depicted. Among the numerous input variables, the focus is placed on the correlation between sensor data. Dark blue indicates positive correlations, light blue represents negative correlations, and white signifies no correlation. Notably, when the row and column numbers are identical, the correlation coefficient is always one, as it denotes the correlation of the feature with itself.

After performing the correlation analysis, a subset of features was selected for each dataset. In the case of the NASA turbofan dataset, only 14 features were chosen for further analysis. Similarly, for the Li-ion battery dataset, a total of 4 features were deemed relevant and included in subsequent modeling efforts. This feature selection process was instrumental in reducing the dimensionality of the datasets, focusing on the most informative sensor data, and improving the overall accuracy of the models.

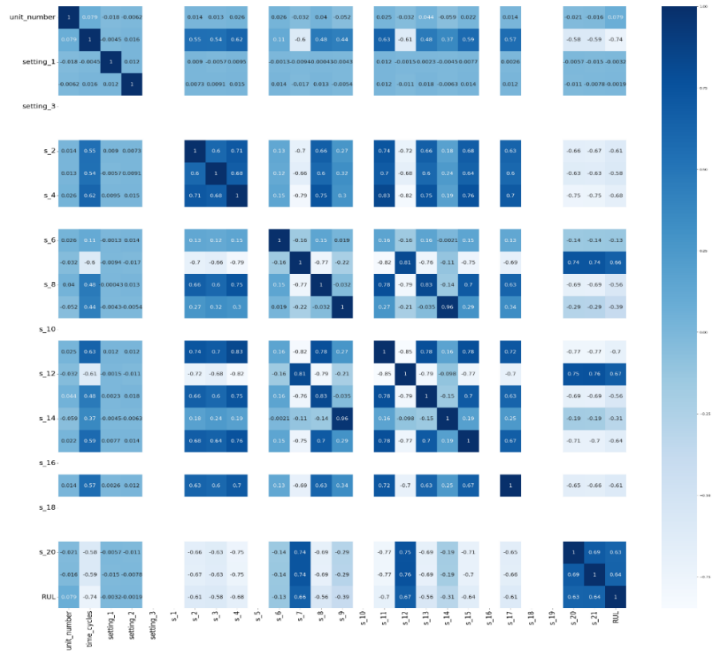

Figure 5

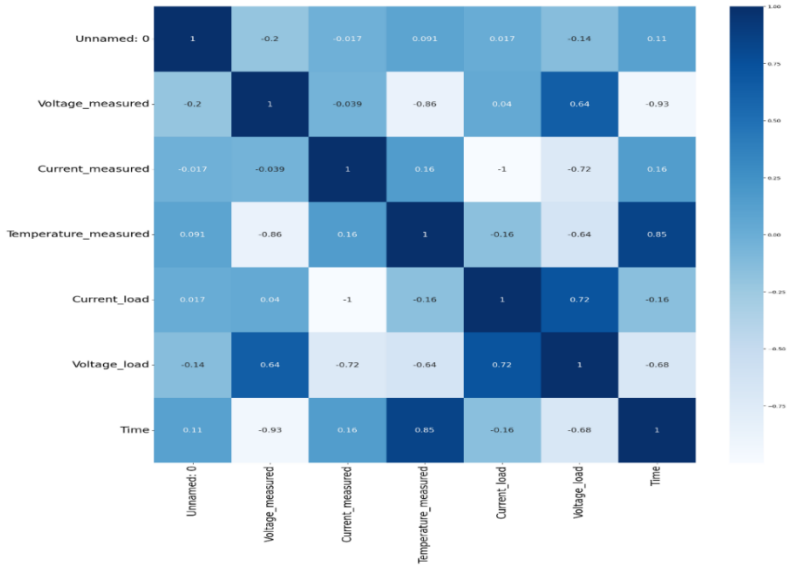

Figure 6

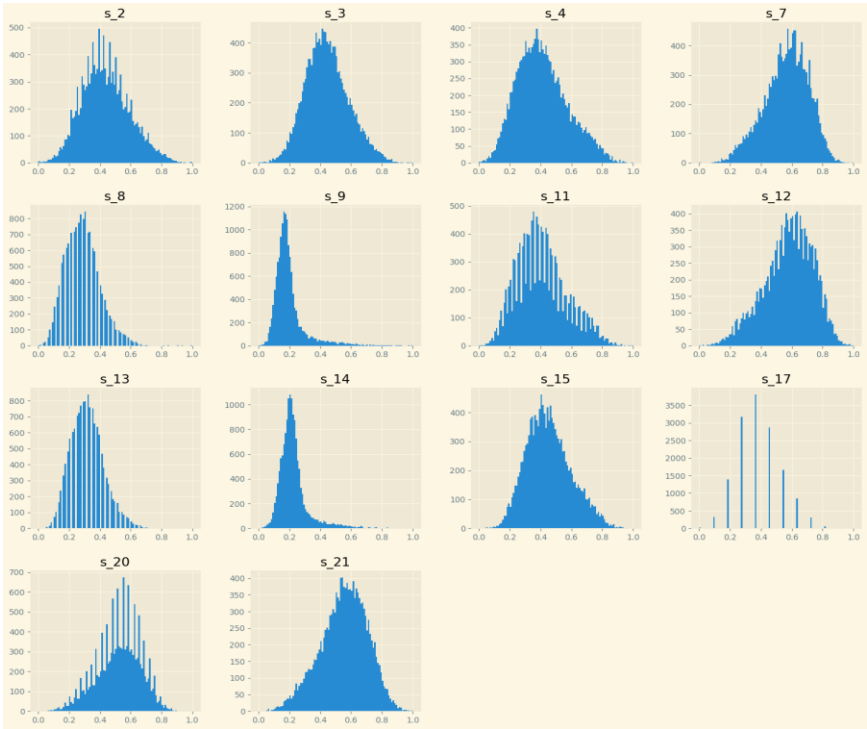

Figure 7

3.1.3 Traditional machine learning algorithms

In the part of traditional machine learning, our analysis consisted of three rounds to investigate the impact of different approaches on the predictive accuracy of the NASA turbofan dataset. Additionally, a single round of analysis was conducted for the Lithium-ion battery dataset, employing various algorithms.

First, I am giving an overview on the algorithms we use, Support Vector Regression, the objective of SVR is to find a hyperplane that best fits the data points in a continuous space. This is achieved by mapping the input variables to a high dimensional feature space and finding a hyperplane that maximizes the margin distance between the hyperplane and the closest data points, while also minimizing the prediction error.

Next, Random Forest regression, in machine learning is an ensemble technique capable of performing both regression and classification tasks with the use of multiple decision trees and a technique called Bootstrap and Aggregation, commonly known as bagging. The basic idea behind this is to combine multiple decision trees in determining the final output rather than relying on individual decision trees.

Next, Linear regression, Linear regression is one of the easiest and most popular Machine Learning algorithms. It is a statistical method that is used for predictive analysis. Linear regression makes predictions for continuous variables and it shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.

finally, XGBoost, or extreme Gradient Boosting, is a machine learning

algorithm under ensemble learning. It is trendy for supervised learning tasks, such as regression and classification. XGBoost builds a predictive model by combining the predictions of multiple individual models, often decision trees, in an iterative manner XGBoost is widely used in various domains due to its high predictive performance and versatility across different datasets

During the first round of analysis for the turbofan dataset, we utilized all available data, including the sensors that were identified as useless. The applied algorithms were Linear Regression, Random Forest Regression, and Support Vector Regression (SVR) using sklearn library. However, the results yielded a modest predictive accuracy, with a test dataset R2 score of 0.57 and a train dataset R2 score of 0.95, indicating room for improvement.

In the second round, we excluded the previously identified useless sensors and repeated the analysis using the same algorithms. This adjustment led to a slight improvement in predictive accuracy, with a test dataset R2 score of 0.64 and a train dataset R2 score of 0.96. Although this was an advancement from the first round, the achieved results were still considered suboptimal.

For the third round of analysis, several significant enhancements were introduced to improve the modeling approach. Aside from excluding irrelevant sensors, time series techniques were incorporated to leverage historical data. A new feature called the 'rolling mean feature' was introduced, which represents the average value of a feature over a period of 10-time cycles. This feature was calculated for each individual feature, and the same algorithms were applied for modeling.

The results of this round showed substantial improvements compared to previous

rounds. The best achieved outcome was an R2 score of 0.752 for the test dataset. This score indicates a higher level of correlation between the predicted values and the actual values, suggesting that the model's predictions align more closely with the true values. Essentially, the model's ability to capture the underlying patterns and relationships in the data has improved significantly.

Furthermore, the model achieved an impressive R2 score of 0.97 for the train dataset. This score indicates a strong performance in capturing the underlying patterns within the training data. It suggests that the model has effectively learned and represented the relationships in the historical data, resulting in highly accurate predictions.

Moving on to the analysis of the Lithium-ion battery dataset, a single round was conducted using all available data, excluding the useless sensors. The XGBoost algorithm was employed, resulting in remarkable accuracy scores. The model achieved a train dataset accuracy of 0.999 and a test dataset accuracy of 0.9972.

3.1.4 Deep Learning algorithms

In addition to the previous analysis, deep learning algorithms were applied to the NASA Turbofan dataset to further explore the prediction of Remaining Useful Life (RUL). Specifically, Neural Network (NN) and Long Short-Term Memory (LSTM) algorithms were employed in this study.

First, I am giving an overview on the algorithms we used, NN, A neural network is a machine learning program, or model, that makes decisions in a manner similar to the human brain, Every neural network consists of layers of nodes, or artificial neurons—an input layer, one or more hidden layers, and an output layer. Each node connects to others, and has its own associated weight and threshold. If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network. Otherwise, no data is passed along to the next layer of the network.

Next, LSTM, is an improved version of recurrent neural network, has a single hidden state that is passed through time, which can make it difficult for the network to learn long-term dependencies. LSTMs address this problem by introducing a memory cell, which is a container that can hold information for an extended period. LSTM networks are capable of learning long-term dependencies in sequential data, which makes them well-suited for tasks such as time series forecasting like predictive maintenance. The memory cell is controlled by three gates: the input gate, the forget gate, and the output gate. These gates decide what information to add to, remove from, and output from the memory cell. The input gate controls what information is added to the memory cell. The forget gate

controls what information is removed from the memory cell. And the output gate controls what information is output from the memory cell. This allows LSTM networks to selectively retain or discard information as it flows through the network, which allows them to learn long-term dependencies. Furthermore, the dataset was preprocessed and features were selected accordingly. The performance of the models was evaluated using appropriate metrics to assess their accuracy and predictive capabilities.

Firstly, for the NN machine learning model, the model architecture comprises three densely connected layers, formulated using the Keras framework. The input layer, consisting of 64 neurons with rectified linear unit (ReLU) activation, accepts data samples with dimensions corresponding to the feature space of the training dataset. Subsequently, two hidden layers, each comprising 64 neurons activated by ReLU, facilitate the extraction of intricate patterns within the data. Finally, the output layer, containing a single neuron, produces continuous predictions. The model is optimized using the Adam optimizer and trained to minimize the mean squared error loss function. On the other hand, two distinct LSTM models are designed and evaluated on a common dataset. The first model, denoted as the "Enhanced LSTM," comprises two LSTM layers with 128 and 64 neurons, respectively, followed by dense layers with 16 and 1 neuron units, each employing rectified linear unit (ReLU) activation functions.

In contrast, the second model, referred to as the "Reduced LSTM," adopts a simplified architecture featuring a single LSTM layer with 32 neurons and subsequent dense layers with 8 and 1 neuron units, also utilizing ReLU activations. Both models are trained using the Adam optimizer to minimize the mean squared

error loss function. Experimental results, obtained through training and validation on a dataset with 14 input features and 10-time steps, reveal distinct performance characteristics of each architecture. The Enhanced LSTM model demonstrates enhanced predictive capabilities attributed to its deeper architecture, while the Reduced LSTM model offers computational efficiency with reduced model complexity. This comparative analysis provides valuable insights into the trade-offs between model complexity and prediction accuracy in LSTM-based time series forecasting. On the other hand, the Li-ion dataset exclusively underwent analysis using the NN algorithm. The dataset was prepared and features were selected to create a machine learning model tailored to the characteristics of the Li-ion battery dataset. The performance of the NN model was assessed to determine its effectiveness in predicting the RUL of the Li-ion batteries. The model architecture comprises three densely connected layers, formulated using the Keras framework. The input layer, consisting of 64 neurons with rectified linear unit (ReLU) activation, accepts data samples with dimensions corresponding to the feature space of the training dataset. Subsequently, two hidden layers, each comprising 64 neurons activated by ReLU, facilitate the extraction of intricate patterns within the data. Finally, the output layer, containing a single neuron, produces continuous predictions. The model is optimized using the Adam optimizer and trained to minimize the mean squared error loss function. This architecture demonstrates potential applicability in diverse domains requiring accurate regression estimations. Experimental evaluations showcase its effectiveness in capturing complex relationships within the data, thereby highlighting its utility in predictive modeling

3.2 Experimental Results

Algorithm	Train dataset r2 score	Test dataset r2 score	Dataset used
Linear regression round 1	**0.5830**	**0.5360**	Nasa turbofan
Linear regression round 2	**0.6502**	**0.5203**	Nasa turbofan
Linear regression round 3	**0.6597**	**0.6368**	Nasa turbofan
Random forest round 1	**0.9506**	**0.5703**	Nasa turbofan
Random forest round 2	**0.9607**	**0.6416**	Nasa turbofan
Random forest round 3	**0.9740**	**0.7513**	Nasa turbofan
XGBoost round 2	**0.6882**	**0.5483**	Nasa turbofan
XGBoost round 3	**0.8379**	**0.7431**	Nasa turbofan
SVR round 1	**0.6057**	**0.5122**	Nasa turbofan
SVR round 2	**0.7064**	**0.4810**	Nasa turbofan
SVR round 3	**0.7138**	**0.6829**	Nasa turbofan
LSTM [Figure 8]	**0.8269**	**0.8034**	Nasa turbofan
LSTM Reduced [Figure 9]	**0.7265**	**0.7146**	Nasa turbofan
Neural network	**0.7165**	**0.7046**	Nasa turbofan

Linear regression	**0.8891**	**0.8816**	Li-ion battery
Random forest [Figure 10]	**0.9960**	**0.9995**	Li-ion battery
SVR	**0.2039**	**0.1972**	Li-ion battery
XGBoost	**0.9999**	**0.9971**	Li-ion battery
Neural Network	**0.9795**	**0.9790**	Li-ion battery

Table 4

As shown in the table 4, firstly for the Nasa turbofan results, the findings reveal that the Random Forest Regression consistently outperforms other models, exhibiting the lowest Root Mean Squared Error (RMSE) and highest coefficient of determination (R2) scores. In contrast, SVR demonstrates relatively weaker performance, particularly in the initial rounds. XGBoost and TensorFlow-based neural network models show competitive performance, with the LSTM TensorFlow model displaying promising results [Figure 8]. However, a simplified version of the LSTM model exhibits a slight decrease in performance compared to its full counterpart [Figure 9]. Overall, the Random Tree Regression emerges as the most reliable model for predictive modeling in this context, offering a balance between computational efficiency and predictive accuracy. The study provides valuable insights into the strengths and limitations of various regression models, aiding in informed decision-making for predictive modeling tasks.

On the other hand, the Li-ion Battery results, the examined models include Linear Regression, Random Tree Regression, Support Vector Regression (SVR), XGBoost, and a TensorFlow-based Neural Network (NN). Performance metrics, specifically root mean squared error (RMSE) and coefficient of determination (R2), are analyzed for both test and training sets across each model. Results reveal that the Random Tree Regression model outperforms others, exhibiting substantially lower RMSE values and notably high R2 scores. In contrast, SVR demonstrates poorer performance, evidenced by higher RMSE values and relatively lower R2 scores, indicating its limited suitability for this application. The XGBoost model showcases remarkable predictive accuracy, with significantly low RMSE and high R2 scores on both test and training sets [Figure 10], underscoring its effectiveness for Lithium-Ion battery performance prediction. Additionally, the TensorFlow-based NN model delivers competitive performance, maintaining high R2 scores despite slightly higher RMSE values compared to XGBoost. While Linear Regression demonstrates respectable performance, it falls short compared to Random Tree Regression, XGBoost, and the TensorFlow-based NN model. These findings offer valuable insights into selecting appropriate regression models for Lithium-Ion battery performance prediction, facilitating advancements in renewable energy technology.

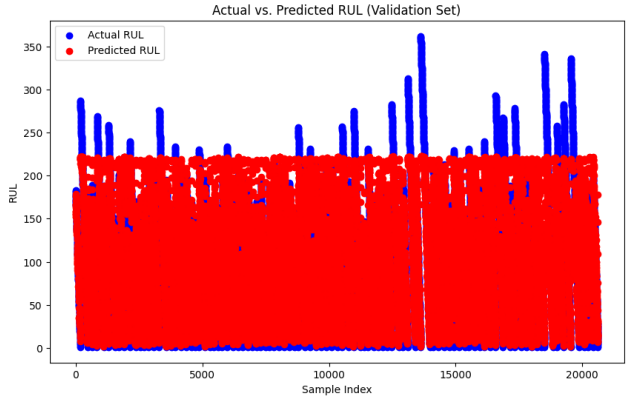

Figure 8

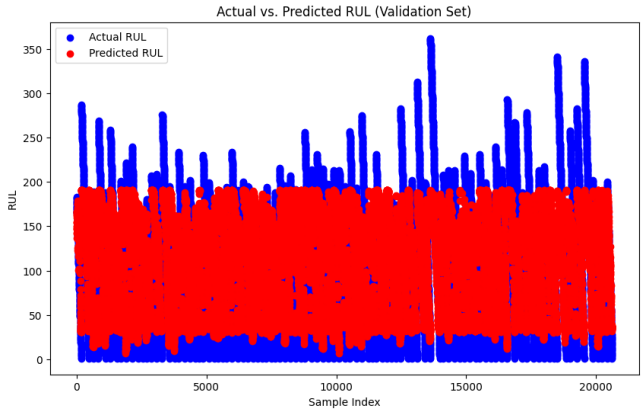

Figure 9

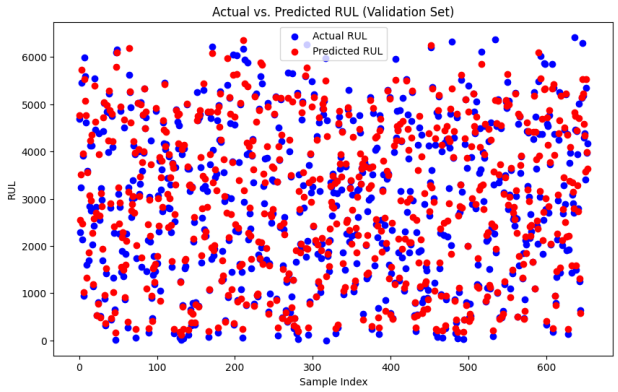

Figure 10

3.3 Module Implementation

The emergence of edge computing has revolutionized the landscape of predictive maintenance, enabling real-time analysis and decision-making at the point of data generation. This section focuses on embedding LSTM and NN machine learning models, developed using TensorFlow, into the ESP8266 module [Figure 11] for predictive maintenance applications. By harnessing the computational power of embedded systems, the ESP8266 module serves as a decentralized processing unit capable of executing complex machine learning algorithms locally, thereby minimizing latency and bandwidth requirements associated with cloud-based solutions.

The integration process starts with converting the LSTM and NN models from TensorFlow-to-TensorFlow Lite format, which is specifically designed for deployment on resource-constrained devices. This conversion helps optimize the models for efficient execution on such devices.

Once the models are in TensorFlow Lite format, they can be transformed into C code using tools like GIT bash. This step ensures compatibility with the ESP8266 microcontroller architecture, which is the target platform for deployment. By converting the models to C code, they can be directly executed on the ESP8266 without relying on the TensorFlow framework.

During this process, several considerations are taken into account to optimize the model size, computational efficiency, and memory utilization. Techniques such as model layers reduction and tuning the number of epochs is employed to reduce the size of the LSTM model. By reducing the number of layers, the overall complexity

of the model is reduced, resulting in a smaller model size. Tuning the number of epochs helps strike a balance between training the model sufficiently and preventing overfitting, which can lead to a larger model size.

Furthermore, feature engineering techniques are applied to enhance the performance of the models. Feature engineering involves transforming or creating new features from the existing data to capture more relevant information. This process helps improve the model's ability to extract meaningful patterns and make accurate predictions.

The successful integration of LSTM and NN models into the ESP8266 module enables real-time prediction of machinery condition based on sensor data inputs. [Figures 12-14] showcase the functionality and efficiency of the embedded models, demonstrating their potential for predictive maintenance applications in industrial settings. The compact size and low computational requirements of the integrated models make them suitable for deployment in edge computing environments, offering autonomous and decentralized predictive maintenance capabilities.

Figure 11

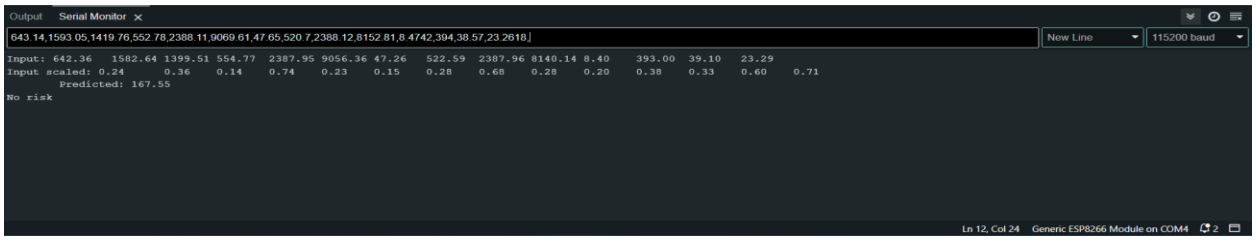

Figure 12

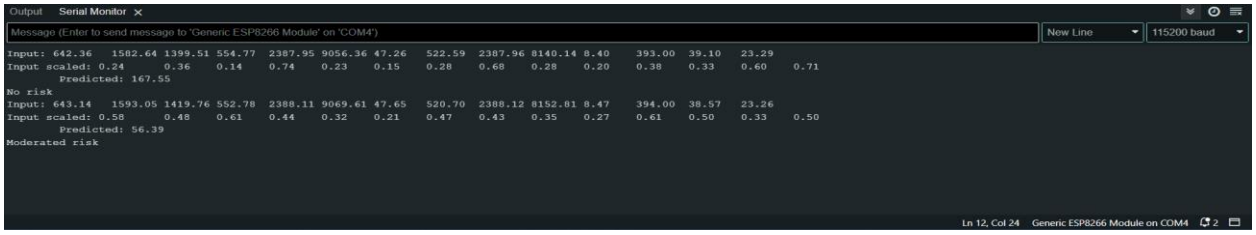

Figure 13

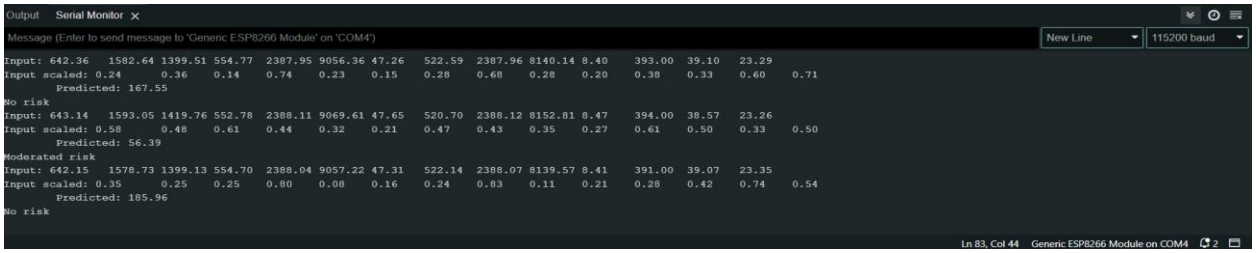

Figure 14

3.4 Android Application

In the development process within Android Studio, an Android application was created for the predictive maintenance tasks for machinery, specifically targeting NASA turbofan engines and Li-ion batteries. A deep learning machine learning model was trained using tensor flow library and deployed to predict machinery condition based on user-input sensor data. The chosen model was then converted to TensorFlow Lite format, which enables efficient deployment on mobile devices. This conversion process involved optimizing the model's size and performance while preserving its predictive capabilities. The TensorFlow Lite model was integrated into an Android application using Android Studio. The user interface was designed to provide a seamless experience for users, allowing them to input sensor data and receive real-time predictions on the machine's condition. The app's functionality was thoroughly tested to ensure accurate predictions and a user-friendly interface as shown in the application [Figures 15 - 26].Within the application, a dedicated section provides users with a concise yet informative overview of the machinery and battery components, by consolidating these details into a user-friendly interface, the application enables users to quickly grasp the essential characteristics of the machinery, facilitating efficient decision-making processes and informed maintenance strategies. This overview section serves as a valuable resource for users seeking a comprehensive understanding of the machinery's capabilities and operational parameters, contributing to improved management and optimization of machinery performance.

Figure 15

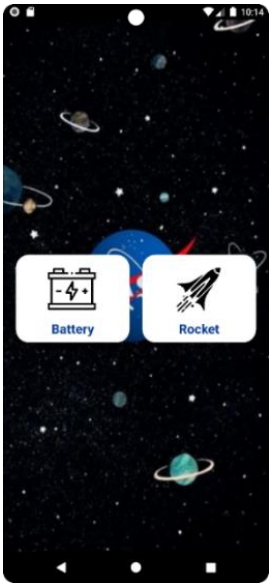

Figure 16

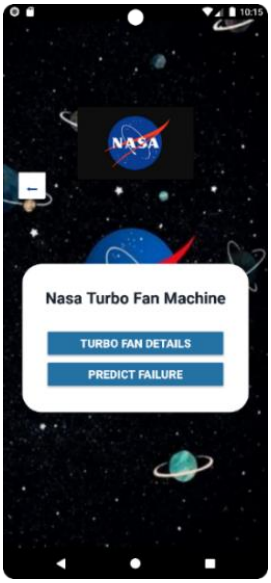

Figure 17

Figure 18

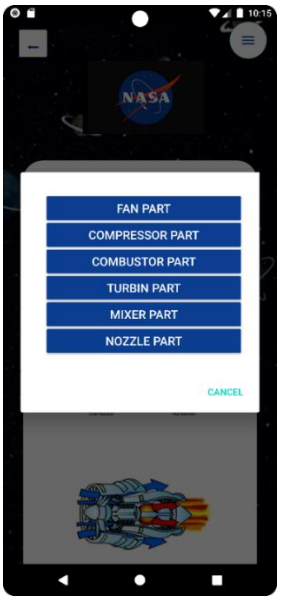

Figure 19

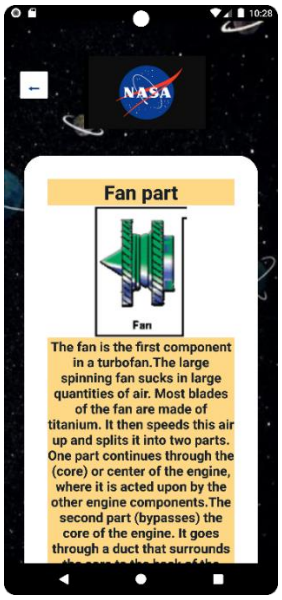

Figure 20

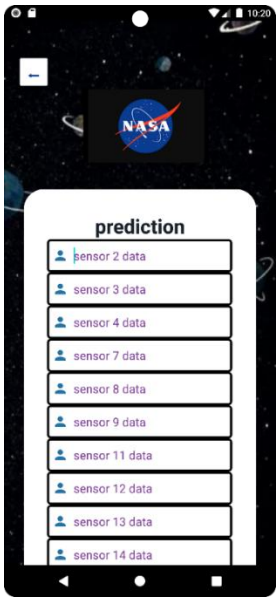

Figure 21

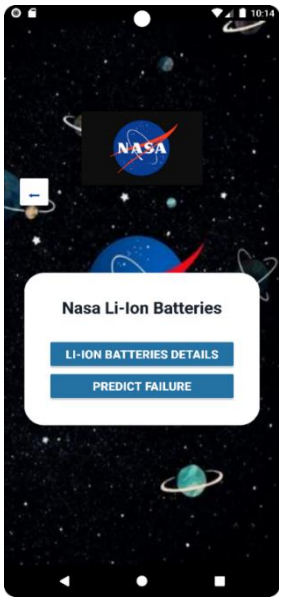

Figure 22

Figure 23

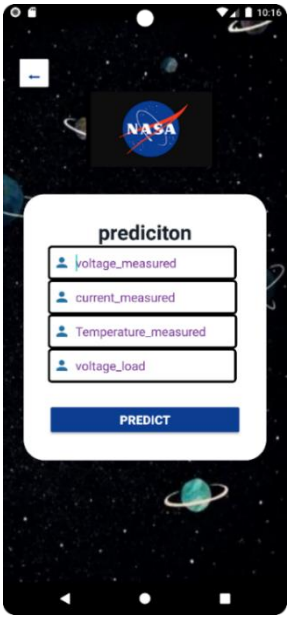

Figure 24

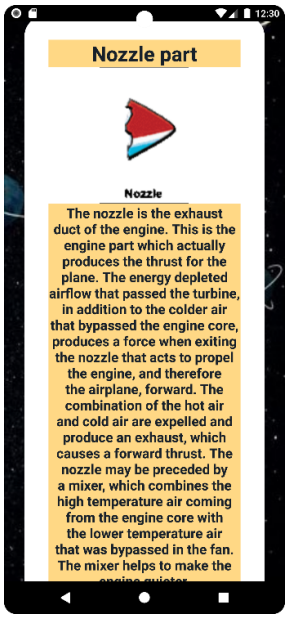

Figure 25

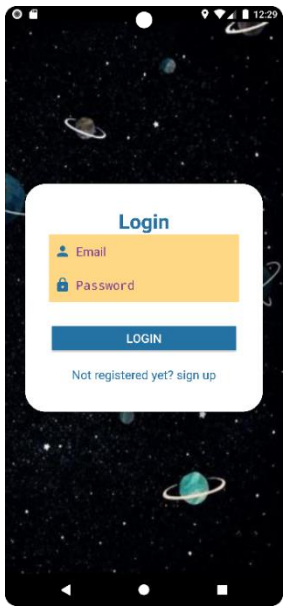

Figure 26

Chapter 4

Conclusion

In this thesis, we conducted a comprehensive study focusing on the prognostication of the Remaining Useful Life (RUL) for turbofan engines and NASA Li-ion battery datasets. The research aimed to investigate the performance of both traditional machine learning algorithms and deep learning algorithms in predicting the risk of failure. Additionally, preprocessing steps were employed to enhance the quality of the input data, including correlation analysis, feature engineering, and scaling.

The field of predictive maintenance, driven by Artificial Intelligence (AI) techniques, has gained significant importance in recent years. By leveraging advanced data analytics and machine learning algorithms, predictive maintenance aims to optimize maintenance activities, reduce downtime, and enhance the reliability of industrial systems. Our study contributes to this area by specifically focusing on AI-driven predictive maintenance techniques for turbofan engines and Li-ion batteries.

The literature review revealed the effectiveness of AI techniques, such as support vector machines (SVMs), random forests, and artificial neural networks, in predictive maintenance. These algorithms have been widely employed for fault detection, diagnosis, and RUL prediction, utilizing historical data to learn patterns and make accurate predictions. However, there is a lack of research specifically targeting the RUL prediction of turbofan engines and Li-ion batteries.

In our methodology, we utilized the NASA Turbofan dataset and conducted three rounds of analysis, as well as a single round of analysis for the Li-ion dataset. The analysis involved evaluating different feature selections and the inclusion of historical data to identify patterns and improve prediction accuracy. Traditional machine learning algorithms, including linear regression, support vector regression (SVR), and random tree regression, were applied in the first round. Deep learning algorithms, such as neural networks (NN) and long short-term memory (LSTM), were employed in subsequent analysis for the NASA Turbofan dataset.

The results shows that the random tree regression consistently outperformed other models for the NASA Turbofan dataset, exhibiting the lowest root mean squared error (RMSE) and the highest coefficient of determination (R2) scores. SVR showed relatively weaker performance, particularly in the initial rounds. XGBoost and TensorFlow-based neural network models displayed competitive performance, with the LSTM TensorFlow model showing promising results. For the Li-ion battery dataset, the random tree regression model outperformed others, exhibiting substantially lower RMSE values and notably high R2 scores. XGBoost also showcased remarkable predictive accuracy for Li-ion battery performance prediction.

For the future work, while this thesis provides valuable insights into the prognostication of Remaining Useful Life (RUL) for turbofan engines and NASA Li-ion battery datasets using AI-driven predictive maintenance techniques, there are several avenues for future research and improvement. The following are some potential areas for future work:

1. Exploration of advanced deep learning architectures: In this study, we employed basic deep learning architectures such as neural networks (NN) and long short-term memory (LSTM). Future research can explore more advanced architectures, such as convolutional neural networks (CNN), recurrent neural networks (RNN), or transformer-based models, to further enhance the predictive capabilities. These architectures may capture complex temporal dependencies and spatial patterns in the data, leading to improved accuracy.

2. Ensemble learning approaches: Ensemble learning techniques, such as stacking, bagging, and boosting, can be investigated to combine the predictions of multiple models and improve overall performance. By leveraging the strengths of different algorithms, ensemble models can provide more robust and accurate predictions.

3. Integration of domain knowledge: Incorporating domain knowledge into the predictive models can enhance their interpretability and domain-specific performance. The inclusion of expert knowledge, physical models, and domain-specific features can improve the accuracy of RUL prediction and provide valuable insights into the underlying mechanisms of failure.

4. Real-time implementation and deployment: To translate the research findings into practical applications, future work should focus on the real-time

implementation and deployment of the predictive maintenance models. Developing efficient algorithms and architectures that can handle streaming data and provide timely predictions will be crucial for effective maintenance planning and decision-making.

5. Validation on diverse datasets: While this study utilized the NASA Turbofan and Li-ion battery datasets, future research should validate the proposed models on diverse datasets from different sources and industries. This will help assess the generalizability and robustness of the developed models and ensure their applicability in real-world scenarios.

6. Integration with condition monitoring systems: Integrating the predictive maintenance models with real-time condition monitoring systems can enable continuous monitoring, early fault detection, and proactive maintenance. Future work should focus on developing frameworks and interfaces for seamless integration with existing industrial monitoring systems.

In conclusion, future research in the field of predictive maintenance for turbofan engines and Li-ion batteries should explore advanced deep learning architectures, ensemble learning approaches, and the integration of domain knowledge. Further advancements in feature selection and engineering, uncertainty estimation, real-time implementation, and validation on diverse datasets are also crucial. The integration with condition monitoring systems will enhance the practical applicability of the developed models, leading to improved maintenance planning and system reliability.

In summary, our study provides valuable insights into the strengths and limitations of various regression models for predictive maintenance of turbofan engines and Li-ion batteries. The random tree regression model emerges as the most reliable model, offering a balance between computational efficiency and predictive accuracy. The findings contribute to informed decision-making for predictive modeling tasks in the context of these specific applications.

These results have significant implications for the field of renewable energy technology, as accurate RUL prediction for Li-ion batteries can optimize their performance and lifespan, leading to improved efficiency and reliability in renewable energy systems.

Overall, this thesis contributes to the advancement of AI-driven predictive maintenance techniques for turbofan engines and Li-ion batteries. The research highlights the importance of selecting appropriate regression models based on their performance metrics and offers valuable insights for future research and practical applications in the field of predictive maintenance.

References

1] Babu, G.S.; Zhao, P.; Li, X.L. Deep convolutional neural network-based regression approach for estimation of remaining useful life. In International Conference on Database Systems for Advanced Applications; Springer: Cham, Switzerland, 2016; pp. 214–228.

[2] Zhang, C.; Lim, P.; Qin, K.; Tan, K.C. Mult objective Deep Belief Networks Ensemble for Remaining Useful Life Estimation in Prognostics. IEEE Trans. Neural Network. Learn. Syst. 2016, 28, 2306–2318.

[3] Malhotra, P.; Vishnu, T.V.; Ramakrishnan, A.; Anand, G.; Vig, L.; Agarwal, P.; Shroff, G. Multisensory prognostics using an unsupervised health index based on LSTM encoder-decoder. arXiv 2020, arXiv:1608.06154.

[4] Aggarwal, K.; Atan, O.; Farahat, A.K.; Zhang, C.; Ristovski, K.; Gupta, C. Two Birds with One Network: Unifying Failure Event Prediction and Time-to-Failure Modeling. In Proceedings of the 2018 IEEE International Conference on Big Data (Big Data), Seattle, WA, USA, 10–13 December 2018; pp. 1308–1317.

[5] Zheng, S.; Ristovski, K.; Farahat, A.; Gupta, C. Long Short-Term Memory Network for Remaining Useful Life Estimation. In Proceedings of the 2017 IEEE International Conference on Prognostics and Health Management (ICPHM), Ottawa, Canada, 19–21 June 2017; pp. 88–95.

[6] Song, Y.; Xia, T.; Zheng, Y.; Zhuo, P.; Pan, E. Remaining useful life prediction of turbofan engine based on Autoencoder-BLSTM.Comput. Integr. Manuf. Syst. 2019, 25, 1611–1619.

[7] Wen, L.; Li, X.; Gao, L.; Zhang, Y. A new convolutional neural network-based data-driven fault diagnosis method. IEEE Trans. Ind. Electron. 2018, 65, 5990–5998.

[8] Li, J.; Li, X.; He, D. A directed acyclic graph network combined with CNN and LSTM for remaining useful life prediction. IEEE Access 2019, 7, 7546475 475.

[9] Muneer, A.; Taib, S.M.; Fati, S.M.; Alhussian, H. Deep-Learning Based Prognosis Approach for Remaining Useful Life Prediction of Turbofan Engine. Symmetry 2021, 13, 1861.

[10] Zhang, X.; Xiao, P.; Yang, Y.; Cheng, Y.; Chen, B.; Gao, D.; Liu, W.; Huang, Z. Remaining Useful Life Estimation Using CNN-XGB With Extended Time Window. IEEE Access 2019, 7, 154386–154397.

[11] Ji, S.; Han, X.; Hou, Y.; Song, Y.; Du, Q. Remaining Useful Life Prediction of Airplane Engine Based on PCA–BLSTM. Sensors 2020,20, 4537.

[12] Wang, R.; Shi, R.; Hu, X.; Shen, C. Remaining Useful Life Prediction of Rolling Bearings Based on Multiscale Convolutional Neural Network with Integrated Dilated Convolution Blocks. Shock Vib. 2021, 1–11.

[13] A. Kanys, “An ontology-based knowledge modelling for a sustainability assessment domain,” *Sustainability*, vol. 10, no. 2, p. 300, 2018.

[14] V. Mathew, T. Toby, V. Singh, B. M. Rao, and M. G. Kumar, “Prediction of remaining useful lifetime (RUL) of turbofan engine using machine learning,” in *2017 IEEE International Conference on Circuits and Systems (ICCS)*, pp. 306–311, Dec 2017.

[15] Z. Zheng, J. Peng, K. Deng, K. Gao, H. Li, B. Chen, Y. Yang, and Z. Huang, "A novel method for lithium-ion battery remaining useful life prediction using time window and gradient boosting decision trees," in *2019 10th International Conference on Power Electronics and ECCE Asia (ICPE 2019-ECCE Asia)*, pp. 3297–3302. IEEE, 2019.

[16] A. Bakir, M. Zaman, A. Hassan, and M. Hamid, "Prediction of remaining useful life for mech equipment using regression," in *Journal of Physics: Conference Series*, vol. 1150, no. 1, p. 012012. IOP Publishing, 2019.

[17] W. Mao, J. Chen, X. Liang, and X. Zhang, "A new online detection approach for rolling bearing incipient fault via self-adaptive deep feature matching," *IEEE Transactions on Instrumentation and Measurement*,2019.

[18] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *nature*, vol. 521, no. 7553, p. 436, 2015.

[19] Y. LeCun, B. E. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. E. Hubbard, and L. D. Jackel, "Handwritten digit recognition with a back propagation network," in *Advances in neural information processing systems*, pp. 396–404, 1990.

[20] Wang, J.; Wen, G.; Yang, S.; Liu, Y. Remaining useful life estimation in prognostics using deep bidirectional lstm neural network. In Proceedings of the 2018 Prognostics and System Health Management Conference (PHM-Chongqing), Chongqing, China, 26–28 October 2018; pp. 1037–1042.